Name

- For the ODE $y' = A\,y + \begin{pmatrix} e^{2t} \\ 3\,e^{2t} \end{pmatrix}$ with A below

In[ ]:=
```
MatrixForm[Eigensystem[(2 1
                        1 2)]]
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Out[ ]//MatrixForm=

$$\begin{pmatrix} 3 & 1 \\ \{1,1\} & \{-1,1\} \end{pmatrix}$$

- Write down the gen sol.

$y_c(t) =$

- Write down the form for the particular solution

$y_p(t) =$

- Write down the Lin System for the coefficients

1. For the ODE $y' = A\,y + \begin{pmatrix} e^{2t} \\ -e^{2t} \end{pmatrix}$ where $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

1.1. Write down the form for the particular solution

$y_p(t) =$

1.2. Write down the Lin System for the coefficients

- For the ODE $y' = A\,y + \begin{pmatrix} \sin(2\,t) \\ -\sin(2\,t) \end{pmatrix}$ where $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.
  - Write down the form for the particular solution

    $y_p(t) =$

  - Write down the Lin System for the coefficients

2. For the ODE $y' = A\,y + \begin{pmatrix} t+1 \\ 2\,t \end{pmatrix}$ where $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

2.1. Write down the form for the particular solution

$y_p(t) =$

2.2. Write down the Lin System for the coefficients